

KOMPRESS v2: Geometric-Mean Consensus Distillation, Spectral Rigidity, and Structural AST Invariants for Ultra-Low-Latency Edge Reasoning

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ABSTRACT

We introduce **KOMPRESS v2**, a mathematically grounded framework for distilling and compressing frontier language models into ultra-low-latency 1.7B edge student models with ternary weights ($\{-1, 0, +1\}$). KOMPRESS v2 provides three central contributions: (1) **Consensus Equivalence in Multi-Teacher Distillation**: we prove that arithmetic averaging of teacher logit vectors induces an exact normalized geometric-mean probability distribution, and that minimizing the mean reverse Kullback-Leibler divergence to individual teachers is optimization-equivalent to minimizing reverse KL divergence against this geometric-mean consensus; (2) **Spectral Rigidity & Restricted Isometry Null Models**: we establish asymptotic Marchenko-Pastur bulk and extreme singular value limits for sparse ternary projection ensembles, identifying the bounded spectral window $0 < c \leq s_i(W|_S) \leq C < \infty$ required on active semantic subspaces to prevent representation explosion and dimensional collapse; and (3) **Structural Invariant Verification**: we formulate a four-predicate admissibility contract $\mathcal{A}(c) = \mathcal{P}(c) \wedge \mathcal{S}(c) \wedge \mathcal{T}(c) \wedge \mathcal{G}(c)$ across isolated git worktrees, decomposing reasoning fidelity into predictive, structural, and behavioral components. Empirically, KOMPRESS v2 achieves a validation cross-entropy of **1.6120** on edge reasoning benchmarks (improving from 2.1369 base and 1.8166 single-teacher KD), reduces resident memory from 3.4 GB to **0.42 GB**, accelerates Apple Silicon M-series inference from 28 to **118 tokens/s**, and yields a 100% verification pass rate across deterministic algorithmic test suites.

1. Introduction

Compressing high-parameter frontier reasoning models into edge-deployable form factors requires balancing three distinct dimensions of fidelity: *predictive fidelity* $\mathcal{F}_{\text{pred}}$ (distributional alignment with

supervisory models), *structural fidelity* $\mathcal{F}_{\text{struct}}$ (syntactic and invariant preservation of generated programs), and *behavioral fidelity* $\mathcal{F}_{\text{behav}}$ (task completion on target domains).

Standard distillation techniques typically supervise a student model with a single teacher under KL divergence or train ternary networks without formal spectral bounds. In this paper, we formalize the mathematical mechanisms underlying multi-faculty distillation, analyze ternary operators through random matrix null models, and verify edge execution via isolated worktree harnesses.

2. Consensus Distillation Theory

2.1 Logit Averaging Induces Geometric-Mean Probability Consensus

Let $\mathcal{V}$ denote the shared token vocabulary ($|\mathcal{V}| = V$), and let $F = \{f_1, f_2, \dots, f_K\}$ denote the Council of Elders (a faculty of K distinct teacher models). For a given context x , each teacher outputs a logit vector $z_k(x) \in \mathbb{R}^V$.

Proposition 1 (Consensus Equivalence)

Let teacher probability distributions at temperature $T > 0$ be defined by $p_k(i) = \frac{\exp(z_{k,i}/T)}{\sum_{j=1}^V \exp(z_{k,j}/T)}$.

The softmax over the arithmetic mean logit vector $\bar{z} = \frac{1}{K} \sum_{k=1}^K z_k$ satisfies:

$$\text{softmax}\left(\frac{\bar{z}}{T}\right)_i = \frac{\left(\prod_{k=1}^K p_k(i)\right)^{1/K}}{\sum_{j=1}^V \left(\prod_{k=1}^K p_k(j)\right)^{1/K}} = \bar{p}(i)$$

That is, logit averaging induces the normalized geometric-mean probability distribution of the faculty.

Proposition 2 (Optimization Equivalence of Mean Reverse KL)

Let q denote the student probability distribution. Minimizing the average reverse KL divergence across individual teachers is optimization-equivalent to minimizing reverse KL divergence against the normalized geometric-mean consensus $\bar{p}$:

$$\frac{1}{K} \sum_{k=1}^K D_{\text{KL}}(q \| p_k) = D_{\text{KL}}(q \| \bar{p}) - \log Z$$

where $Z = \sum_{j=1}^V \left(\prod_{k=1}^K p_k(j) \right)^{1/K}$ is a student-independent normalization constant.

Consequently:

$$\arg \min_q \frac{1}{K} \sum_{k=1}^K D_{\text{KL}}(q \| p_k) = \arg \min_q D_{\text{KL}}(q \| \bar{p})$$

Consensus Mode Suppression: Because reverse KL divergence $D_{\text{KL}}(q \| \bar{p})$ penalizes assigning probability mass where the supervisory target $\bar{p}(i) \approx 0$, and $\bar{p}(i) \propto \prod_{k=1}^K p_k(i)^{1/K}$, any token candidate i assigned near-zero probability by even a single teacher ($p_m(i) \rightarrow 0$) is strongly suppressed in the consensus target ($\bar{p}(i) \rightarrow 0$). This mechanism enforces conservative agreement and suppresses teacher-specific idiosyncratic modes.

3. Spectral Rigidity & Random Matrix Null Models

3.1 Marchenko-Pastur Asymptotics for Ternary Ensembles

We consider a ternary random matrix null ensemble $W \in \{-1, 0, +1\}^{m \times n}$ with independent entries satisfying $\mathbb{P}(W_{ij} = +1) = \mathbb{P}(W_{ij} = -1) = \frac{p}{2}$ and $\mathbb{P}(W_{ij} = 0) = 1 - p$, such that $\mathbb{E}[W_{ij}] = 0$ and $\text{Var}(W_{ij}) = p$.

Theorem 1 (Spectral Edges for Sparse Ternary Matrices)

Let $m, n \rightarrow \infty$ with aspect ratio $\gamma = m/n \in (0, 1]$. The empirical spectral distribution of the sample covariance matrix $\frac{1}{n} W W^T$ converges almost surely to the Marchenko-Pastur distribution μ_{MP} on the support $[\lambda_-, \lambda_+]$:

$$\lambda_{\pm} = p(1 \pm \sqrt{\gamma})^2$$

and the extreme singular values satisfy:

$$\frac{s_{\max}(W)}{\sqrt{n}} \xrightarrow{\text{a.s.}} \sqrt{p}(1 + \sqrt{\gamma}), \quad \frac{s_{\min}(W)}{\sqrt{n}} \xrightarrow{\text{a.s.}} \sqrt{p}(1 - \sqrt{\gamma})$$

Subspace Separability vs. Operator Norm Control: An upper bound $s_{\max}(W) \leq C$ prevents representation explosion, but does not prevent dimensional collapse. To guarantee pairwise distance preservation, a trained BitLinear layer must satisfy restricted lower and upper singular bounds on the active semantic representation subspace $\mathcal{S}$:

$$0 < c \cdot \|x - y\|_2 \leq \|W(x - y)\|_2 \leq C \cdot \|x - y\|_2, \quad \forall x, y \in \mathcal{S}$$

We define the layer-wise spectral rigidity deviation R_ℓ relative to the Marchenko-Pastur null model:

$$R_\ell = \max \left\{ \left| \frac{s_{\max}(W_\ell)^2}{n} - \lambda_+^{(\ell)} \right|, \left| \frac{s_{\min}(W_\ell)^2}{n} - \lambda_-^{(\ell)} \right| \right\}$$

4. Structural Invariant Verification

To evaluate behavioral correctness without relying solely on language model self-evaluation, we define an explicit four-predicate admissibility contract:

$$\mathcal{A}(c) = \mathcal{P}(c) \wedge \mathcal{S}(c) \wedge \mathcal{T}(c) \wedge \mathcal{G}(c)$$

where $\mathcal{P}(c)$ verifies syntax parse validity; $\mathcal{S}(c)$ checks structural AST invariants; $\mathcal{T}(c)$ executes deterministic unit tests; and $\mathcal{G}(c)$ enforces isolated git worktree provenance.

5. Empirical Evaluation

5.1 Distillation Setup & Validation Cross-Entropy

Training is performed over our unified dataset ($N = 20,007$ instruction samples + teacher traces) using AdamW ($\text{lr}_{\text{init}} = 2 \times 10^{-4}$, cosine decay to 10^{-5} , $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.01, gradient clipping 1.0). We employ $\alpha = 0.5$, $\beta(e) = \min(0.5, \frac{e}{2} \cdot 0.5)$, and distillation temperature $T = 1.5$.

Configuration	Faculty Teachers	Parameters	Best Val Cross-Entropy
Base Student (Unquantized)	None	1.7B	2.1369
Single-Teacher Distillation	Qwen3-14B ($T = 1.5$)	1.7B	1.8166
KOMPRESS v2 (Council of Elders)	Qwen3-8B + Qwen3-14B ($T = 1.5$)	1.7B	1.6120

5.2 Hardware Benchmarking & Specifications

Benchmarking was executed on Apple M3 Max hardware (128 GB Unified Memory, 16-core CPU / 40-core GPU, macOS 15.3, Metal Shading Language kernels):

- **Resident Memory:** 3.4 GB (FP16 baseline) $\rightarrow$ **0.42 GB** ($8.1\times$ reduction, comprising 336 MB packed ternary codes, 28 MB group-64 scales, and 56 MB non-quantized RMSNorm/embeddings).
- **Inference Throughput:** 28 tokens/s $\rightarrow$ **118 tokens/s** ($4.2\times$ speedup).
- **AST Verification:** 100.0% pass rate observed across deterministic algorithmic benchmark suites (0% reward-hacking escapes under the verifier test suite).

6. Conclusion

KOMPRESS v2 unifies multi-teacher geometric-mean consensus distillation, random matrix spectral rigidity null models, and multi-agent AST worktree verification into a principled edge reasoning framework. By proving the exact equivalence between logit averaging and geometric-mean probability consensus, we provide a solid mathematical foundation for sovereign, ultra-low-latency edge intelligence.

References

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